

VIRILE AGITUR



KNOX GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT

2001
TRIAL HSC EXAMINATION

Mathematics

- General Instructions
- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided on page 12
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1–10
- All questions are of equal value
- Use a SEPARATE Writing Booklet for each question

RJBD

Total marks (120)
Attempt questions 1 – 10
All questions are of equal value

Answer each question in a SEPARATE Writing Booklet. Extra Writing Booklets are available.

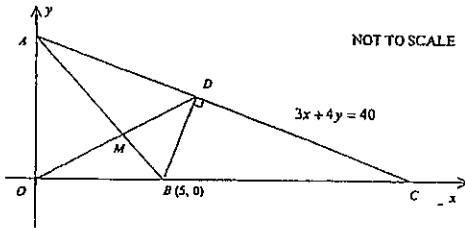
Question 1 (12 marks) Use a SEPARATE Writing Booklet.	Marks
(a) Find the value of $(2^{1.7} + 3)^2$ correct to two decimal places.	2
(b) Solve $4 - 3x > 10$ and graph the solution on the number line.	2
(c) Convert 300° to radians in terms of π .	1
(d) From a pack of fifty-two playing cards a person selects two cards and places them in a row. What is the probability that the second card selected is the ace of hearts?	1
(e) Solve for x : $\frac{2x}{3} - \frac{x+1}{4} = 1$.	2
(f) Find integers a and b such that $\frac{3}{1-\sqrt{2}} = a - \sqrt{b}$.	2
(g) Solve $3^{x-3} = \frac{1}{9}$.	2

NAME: _____ TEACHER: _____

99

Question 2 (12 marks) Use a SEPARATE Writing Booklet.

Marks



The diagram shows the point A on the y -axis and the points $B(5, 0)$ and C on the x -axis. Point D lies on AC such that BD is perpendicular to AC . The equation of the line AC is $3x + 4y = 40$. Point M is the mid-point of OD .

- | | | |
|-----|--|---|
| (a) | Show that A has coordinates $(0, 10)$. | 1 |
| (b) | Find the gradient of the line AB . | 1 |
| (c) | Write down the equation of the line AB . | 1 |
| (d) | Show that the equation of BD is $4x - 3y = 20$. | 1 |
| (e) | By solving the equations of the lines AC and BD simultaneously, show that D has coordinates $(8, 4)$. | 2 |
| (f) | Find the coordinates of M , the mid-point of OD . | 1 |
| (g) | Find the perpendicular distance of D from the line AB . | 2 |
| (h) | Show that M lies on the line AB . | 1 |
| (i) | Find the area of quadrilateral $AOBD$. | 2 |

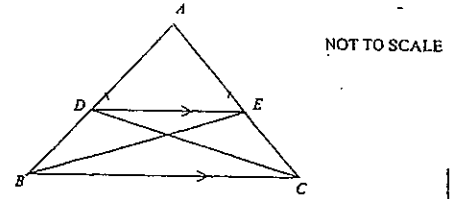
Question 3 (12 marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Differentiate the following:

- | | | |
|-------|-------------------|---|
| (i) | $\sin(3x + 1)$. | 1 |
| (ii) | $x^3 \ln x$. | 2 |
| (iii) | $\frac{e^x}{x}$. | 2 |

- (b)



In the diagram, ABC is an isosceles triangle where $AB = AC$ and DE is parallel to BC .

- | | | |
|------|---|---|
| (i) | Show that ADE is an isosceles triangle. | 2 |
| (ii) | Show that $DB = EC$. | 1 |

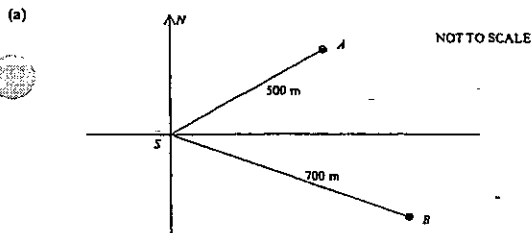
- (c) Find:

- | | | |
|-------|------------------------------|---|
| (i) | $\int (2x + 3)^3 dx$ | 1 |
| (ii) | $\int \sin \frac{x}{2} dx$ | 1 |
| (iii) | $\int_0^1 \frac{dx}{3x + 1}$ | 2 |

100

Question 4 (12 marks). Use a SEPARATE Writing Booklet.

Marks



From a ship at point S two buoys are observed, one at point A at a distance of 500 metres and a bearing of 043° T, the other at point B at a distance of 700 metres and a bearing of 118° T.

- | | | |
|-------|---|---|
| (i) | Copy or trace the diagram into your Writing Booklet and mark on your diagram all the given information. Show $\angle ASB$ is 75° . | 1 |
| (ii) | Find the distance of buoy A from buoy B , correct to the nearest metre. | 2 |
| (iii) | Find the bearing of buoy A from buoy B , correct to the nearest degree. | 3 |
| (b) | The graph of $y = f(x)$ passes through the point $(2, 3)$ and $f'(x) = 3x^2 - 3$. Find an expression for $f(x)$. | 2 |
| (c) | Consider the parabola $(x - 1)^2 = -8(y - 3)$ | |
| (i) | Find the vertex and the focus of the parabola. | 2 |
| (ii) | Sketch the parabola marking the vertex and focus on it. | 1 |
| (iii) | Find the equation of another parabola with the same focal length, focus and axis of symmetry. | 1 |

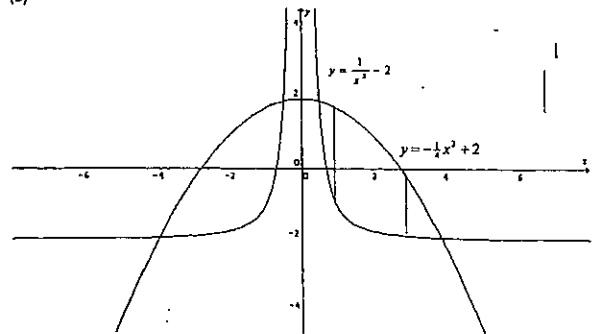
Question 5 (12 marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Consider the curve given by $y = x^3 - 3x^2 + 2$.

- | | | |
|------|--|---|
| (i) | Find the two stationary points and determine their nature. | 4 |
| (ii) | Sketch the curve for $-1 \leq x \leq 3$. | 2 |

- (b)



The diagram shows the graphs of $y = \frac{1}{x^2} - 2$ and $y = -\frac{1}{4}x^2 + 2$.

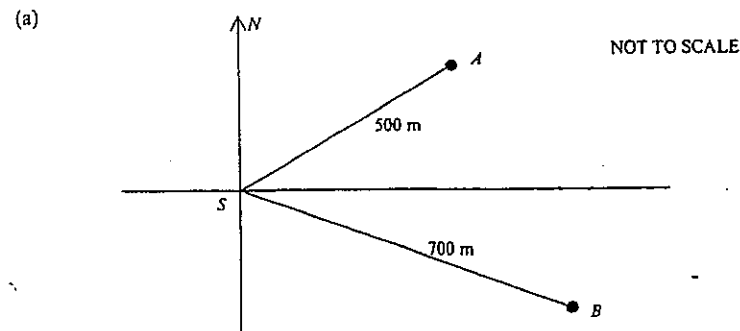
- | | | |
|-------|--|---|
| (i) | Find the area between the curves from $x = 1$ to $x = 3$. | 3 |
| (ii) | Show that the curves intersect when $x^4 - 16x^2 + 4 = 0$. | 1 |
| (iii) | Use the substitution of $u = x^2$, or otherwise, show the x coordinates of the points of intersection of the two curves are | 2 |

$$x = \pm\sqrt{8 \pm 2\sqrt{15}}$$

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Question 4 (12 marks). Use a SEPARATE Writing Booklet.

Marks



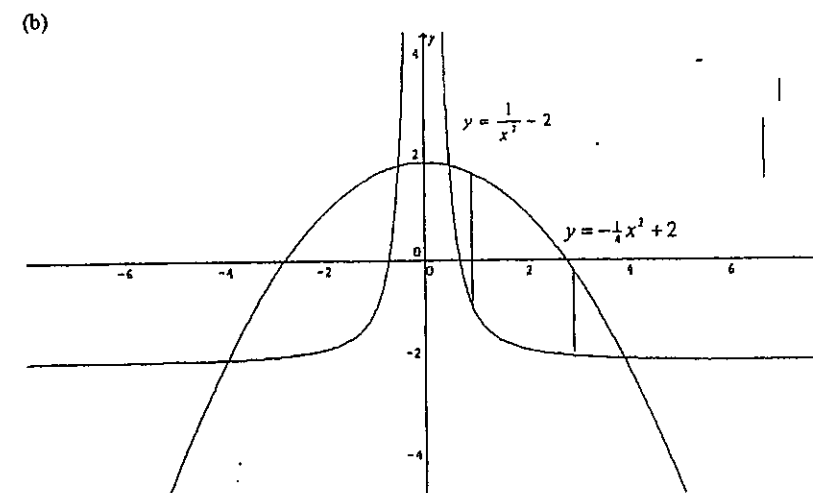
From a ship at point S two buoys are observed, one at point A at a distance of 500 metres and a bearing of 043° T, the other at point B at a distance of 700 metres and a bearing of 118° T.

- | | | |
|--|---|---|
| (i) | Copy or trace the diagram into your Writing Booklet and mark on your diagram all the given information. Show $\angle ASB$ is 75° . | 1 |
| (ii) | Find the distance of buoy A from buoy B, correct to the nearest metre. | 2 |
| (iii) | Find the bearing of buoy A from buoy B, correct to the nearest degree. | 3 |
|
(b) The graph of $y = f(x)$ passes through the point (2, 3) and $f'(x) = 3x^2 - 3$. Find an expression for $f(x)$. | | 2 |
|
(c) Consider the parabola $(x - 1)^2 = -8(y - 3)$ | | |
| (i) | Find the vertex and the focus of the parabola. | 2 |
| (ii) | Sketch the parabola marking the vertex and focus on it. | 1 |
| (iii) | Find the equation of another parabola with the same focal length, focus and axis of symmetry. | 1 |

Question 5 (12 marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Consider the curve given by $y = x^3 - 3x^2 + 2$.
- | | | |
|------|--|---|
| (i) | Find the two stationary points and determine their nature. | 4 |
| (ii) | Sketch the curve for $-1 \leq x \leq 3$. | 2 |



The diagram shows the graphs of $y = \frac{1}{x^2} - 2$ and $y = -\frac{1}{4}x^2 + 2$.

- | | | |
|-------|--|---|
| (i) | Find the area between the curves from $x = 1$ to $x = 3$. | 3 |
| (ii) | Show that the curves intersect when $x^4 - 16x^2 + 4 = 0$. | 1 |
| (iii) | Use the substitution of $u = x^2$, or otherwise, show the x coordinates of the points of intersection of the two curves are | 2 |

$$x = \pm\sqrt{8 \pm 2\sqrt{15}}$$

Question 6 (12 marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Consider the sequence $T_n = 2n + 3$.

- Find the first 3 terms.
- Is this sequence an A.P. or a G.P.? Give reasons.
- Find the sum of the first 20 terms.

1
1
2

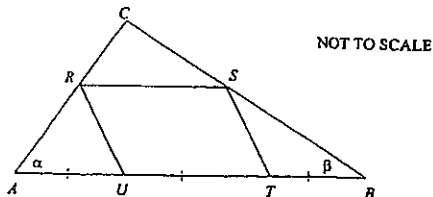
- (b) A pool is being drained and the number of litres of water, L , in the pool at time t minutes is given by the equation:

$$L = 120(40 - t)^2.$$

- At what rate is the water draining out of the pool when $t = 6$ minutes?
- How long will it take for the pool to be completely empty?

2
1

(c)



In the triangle ABC , $\angle CAB = \alpha$, $\angle CBA = \beta$, $AU = UT = TB$ and $RSTU$ is a rhombus. Copy or trace the diagram into your Writing Booklet and mark the information on it.

- Show that $\triangle STB$ is isosceles.
- Show that $\angle STU$ is double the $\angle SBT$.
- Hence prove that $\angle ACB$ is a right angle.

1
1
3

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Question 7 (12 marks). Use a SEPARATE Writing Booklet.

Marks

- (a) Given that $\sin \theta = \frac{3}{\sqrt{15}}$ and $\cos \theta < 0$, find the exact value of $\tan \theta$.

2

- (b) Simplify $\frac{\sin(\frac{\pi}{2} - \theta)}{\sin(\pi - \theta)} \times \tan(-\theta)$.

3

- (c) The points $A(0,4)$, $B(1,4)$, $C(2,0)$, $D(3,-1)$, and $E(4,-1)$ are plotted on a number plane. The line segments joining A to B , B to C , C to D , and D to E , define a function $g(x)$.

- (i) Show clearly on a diagram the information given above.

1

- (ii) Evaluate $\int_0^4 g(x) dx$.

2

- (d) Consider the function $f(x) = \cos^2 x$

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π
$\cos^2 x$				0			

- (i) Copy and complete the above table using exact values.

1

- (ii) Use Simpson's rule with seven function values (from the table), to find an estimation for the volume of the solid formed when the area between the curve $y = \cos x$ and the x -axis between $x = 0$ and $x = \pi$ is revolved about the x -axis.

3

Question 8 (12 marks). Use a SEPARATE Writing Booklet.

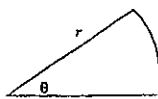
Marks

- (a) A particle is moving in a straight line. Its displacement from a fixed point on the line at time t seconds is given by $x = t - \ln(2t + 1)$, $t \geq 0$, where x is in metres.

- Find an expression for the velocity of the particle.
- Find an expression for the acceleration of the particle and show that the direction of the acceleration is always positive.
- In what direction is the particle moving when $t = 0$?
- At what time does the particle come to rest?
- Where does the particle come to rest?
- How far does the particle travel in the first two seconds of its motion? Leave answer in exact form.

1
1
1
1
1
2

(b)



The sector of a circle with radius r and angle θ has a perimeter of 20 cm.

- Show that the area of this sector can be expressed as $A = 10r - r^2$.
- Find the value of θ , to the nearest degree, which will make this area a maximum.

1
4

103

Question 9 (12 marks) Use a SEPARATE Writing Booklet.

Mark

- (a) (i) Write down an expression for the discriminant of the quadratic equation $ax^2 + 5x + a = 0$

1

- (ii) Hence, find the values of a for which the function $f(x) = ax^2 + 5x + a$ is negative definite.

3

- (b) Solve $\log_2 4 = \log_2 \sqrt{3}$.

2

- (c) The population of soldier ants, S , in a certain area increases exponentially according to $S = Ae^{kt}$, where k is a constant and t is time in weeks.

At the beginning of an observation period there were 5000 ants.

- Calculate the value of the constant A .
- If initially the population of the ants increased at a rate of 350 ants per week, calculate the value of the constant k .
- How many ants will there be after 8 weeks?

1
2
1

- (d) Express in simplest form the limiting sum of the geometric series

2

$$\sin^2 x + \sin^4 x + \sin^6 x + \dots \text{ for } 0 < x < \frac{\pi}{2}.$$

3. a)

$$\frac{d}{dx} (\sin(3x+1)) = 3 \cos(3x+1) \quad (1)$$

$$\frac{d}{dx} (x^2 \ln x) = x^2 \cdot \frac{1}{x} + 2x \ln x \quad (2)$$

$$= x + 2x \ln x$$

$$\frac{d}{dx} \left(\frac{e^x}{x} \right) = \frac{x e^x - e^x (1)}{x^2} \quad (2)$$

$$= \frac{e^x (x-1)}{x^2}$$

) $\triangle ABC$ is isosceles

$$\angle ABC = \angle ACB \quad (\text{base angles})$$

$$\angle ADE = \angle ABC \quad (\text{corresponding angles of } \parallel \text{ lines } DE, BC)$$

$$\angle AED = \angle ACB$$

$$\angle ADE = \angle AED$$

$\triangle ADE$ is isosceles

$$\therefore AD = AE \text{ also } AB = AC$$

$$\text{Hence } AB - AD = AC - AE$$

$$\therefore DB = EC \quad (1)$$

$$(i) \int (2x+3)^5 dx = \frac{1}{2} \times \frac{1}{6} (2x+3)^6 + C$$

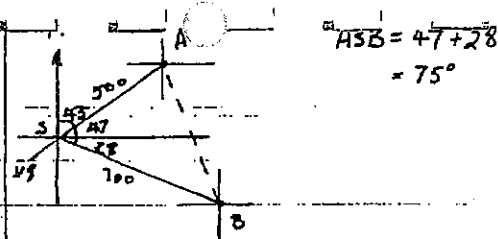
$$= \frac{1}{12} (2x+3)^6 + C \quad (1)$$

$$) \int \sin\left(\frac{x}{2}\right) dx = -2 \cos\left(\frac{x}{2}\right) + C \quad (1)$$

$$i) \int_0^1 \frac{dx}{3x+1} = \left[\frac{1}{3} \ln(3x+1) \right]_0^1$$

$$= \frac{1}{3} \ln(4) - \frac{1}{3} \ln(1)$$

$$= \frac{1}{3} \ln 4 \quad (2)$$



$$(ii) AB^2 = 500^2 + 700^2 - 2(500)(700)\cos 75$$

$$= 740000 - 181173.3316$$

$$= 558826.6684$$

$$AB = 747.547$$

$$= 748 \text{ (nearest metre)} \quad (2)$$

$$(iii) \frac{\sin \angle BBA}{500} = \frac{\sin 75^\circ}{748}$$

$$\sin \angle BBA = 0.64567$$

$$\angle BBA = 40.216$$

$$= 40^\circ \text{ nearest degree}$$

$$\text{Bearing} = 270 + 28 + 40$$

$$= 338^\circ \text{ T.} \quad (3)$$

$$b) f(x) = 3x^2 - 3$$

$$f(x) = x^3 - 3x + c$$

$$3 = (2)^3 - 3(2) + c$$

$$3 = 8 - 6 + c$$

$$c = 1$$

$$f(x) = x^3 - 3x + 1 \quad (2)$$

$$c) (x-1)^2 = -8(y-3)$$

$$(i) \text{ vertex } (1, 3) \quad a = 2$$

$$\text{focus } (-1, 1)$$



$$(iii) \text{ new vertex } (1, -1) \text{ focal length } 2$$

$$(x-1)^2 = 8(y+1) \quad (1)$$

$$(i) \frac{dy}{dx} = 3x^2 - 6x$$

$$\text{S.P. when } \frac{dy}{dx} = 0 \quad 3x(x-2) = 0$$

$$\therefore x = 0 \text{ or } x = 2$$

$$y = 2 \text{ or } y = -2$$

$$\text{S.P. } (0, 2) \quad (2, -2)$$

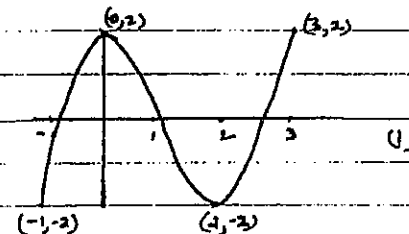
$$\text{nature: } \frac{d^2y}{dx^2} = 6x - 6$$

$$\text{when } x = 0 \quad \frac{d^2y}{dx^2} = -6 < 0 \therefore \text{Concave Down}$$

$$\text{when } x = 2 \quad \frac{d^2y}{dx^2} = 12 - 6 > 0 \therefore \text{Concave up}$$

$$\therefore (0, 2) \text{ max T.P. } (2, -2) \text{ min T.P.}$$

$$(ii) x = -1, y = -2, x = 3, y = 2 \quad (1)$$



$$b) A = \int_1^3 y_T - y_B dx$$

$$= \int_1^3 \left(-\frac{1}{4}x^2 + 2 - x^2 + 2 \right) dx$$

$$= \left[-\frac{1}{12}x^3 + 4x + \frac{1}{x} \right]_1^3$$

$$= \left[\frac{27}{12} + 12 + \frac{1}{3} \right] - \left[-\frac{1}{12} + 4 + 1 \right]$$

$$= 10\frac{1}{12} - 5\frac{1}{12}$$

$$= 5\frac{1}{6}$$

$$(ii) \frac{1}{x^2} - 2 = -\frac{1}{4}x^2 + 2$$

$$4 - 8x^2 = -x^4 + 8x^2$$

$$\therefore x^4 - 16x^2 + 4 = 0$$

$$U = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{16 \pm \sqrt{16^2 - 16}}{2}$$

$$= 8 \pm \frac{4\sqrt{15}}{2}$$

$$\therefore x^2 = 8 \pm 2\sqrt{15}$$

$$x = \pm \sqrt{8 \pm 2\sqrt{15}}$$

$$Q6. a) T_n = 2n + 3$$

$$T_1 = 2(1) + 3 = 5$$

$$T_3 = 2(3) + 3 = 9$$

$$T_5 = 2(5) + 3 = 13$$

$$\text{first 3 terms } 5, 7, 9$$

$$(iii) T_2 - T_1 = 7 - 5 = 2$$

$$T_3 - T_2 = 9 - 7 = 2$$

$$\therefore \text{A.P.}$$

$$(iii) S_{20} = \frac{n}{2} (2a + (n-1)d)$$

$$= \frac{20}{2} (2 \times 5 + 19 \times 2)$$

$$= 10(10 + 38)$$

$$= 480$$

$$b) L = 120(40 - t)^2$$

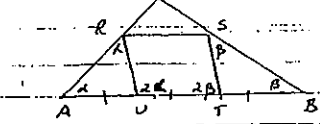
$$(i) \frac{dL}{dt} = -240(40 - t)$$

$$\text{when } t = 6 \quad \frac{dL}{dt} = -240(40 - 6)$$

$$= -8160$$

draining out at 8160 litres per min

(ii) Pool empties in 40 minutes



rhombus has equal sides.
 $UT = ST = TB$

ΔSTB is isosceles

$UT = ST = TB$ (Base angles of isosceles Δ)
 $\angle STU = \angle TSB = \beta$
 $\angle STU = \beta + \beta$ (Exterior angle of a triangle = sum of interior opposite angles)
 $= 2\beta$
 $= 2/58^\circ$

$AU = UT = RU$ (equal sides of rhombus)

ΔAUR is isosceles

$AU = AR = UR = x$ (Base angles of isosceles Δ)
 $\angle URT = \angle UAR$ ext. $\angle$ of $\Delta = \sum$ of interior angles
 $= 2x$
 $= 2/18^\circ$

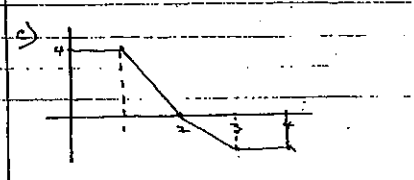
$\angle RUA = \angle STU = 2\beta$ angles of Δ same RU, ST

$\angle URT + \angle RUT = 2\beta + 2x$ (straight angle)
 $= 180^\circ$
 $2(4\beta) = 180$
 $4\beta = 90$

ΔABC
 $\angle ACB + \alpha + \beta = 180$ angle sum of triangle
 $\angle ACB + 90 = 180$
 $\angle ACB = 90$
 ΔABC is a right angle.

$\tan \theta < 0$
 $\tan \theta = -\frac{3}{\sqrt{6}}$
 $\theta = -\frac{1}{2}\sqrt{6}$

$\frac{\sin(\frac{\pi}{2} - \theta)}{\sin(\pi - \theta)} \times \tan(\theta)$
 $= \frac{\cos \theta}{\sin \theta} \times (-\tan \theta)$
 $= -\frac{\cos \theta}{\sin \theta} \times \frac{\sin \theta}{\cos \theta}$
 $= -1$



$\int_0^4 g(x) dx = (4+1) + (\frac{1}{2} \times 4 \times 1) - (\frac{1}{2} \times 1 \times 1) - (1 \times 1)$
 $= 4 + 2 - \frac{1}{2} - 1$
 $= 4\frac{1}{2}$

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π
$\cos^2 x$	1	$\frac{3}{4}$	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{3}{4}$	1
$y_0, y_1, y_2, y_3, y_4, y_5, y_6$	y_0	y_1	y_2	y_3	y_4	y_5	y_6

(ii) $V = \pi \int_0^{\pi} \cos^2 x dx$
 $= \pi \times \frac{1}{2} [y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$
 $= \pi \times \frac{1}{2} [1 + 1 + 4(\frac{3}{4} + 0 + \frac{3}{4}) + 2(\frac{1}{4} + \frac{1}{4})]$
 $= \pi \times \frac{1}{2} [2 + 6 + 1]$
 $= 4.9348 \dots$
 ≈ 4.93 (3dp) units³

i) $\frac{dx}{dt} = 1 - \frac{2}{2t+1}$
 ii) $\frac{d^2x}{dt^2} = \frac{4}{(2t+1)^2} > 0$ since $(2t+1)^2 > 0$
 iii) when $t=0$ $\frac{dx}{dt} = 1 - \frac{2}{1} = -1$

$\therefore$ moves in the negative direction when $t=0$

(iv) rest when $\frac{dx}{dt} = 0$
 $1 - \frac{2}{2t+1} = 0$
 $2t+1 = 2$
 $t = \frac{1}{2}$ Second

(v) at $t = \frac{1}{2}$ $x = \frac{1}{2} - \ln(2 \times \frac{1}{2} + 1)$
 $= \frac{1}{2} - \ln 2$

comes to rest at $\frac{1}{2} - \ln 2 \approx 0.193$ (4dp)

(vi) $x = \frac{1}{2} - \ln 2$ at $t=0$
 $x = 2 - \ln 5$ at $t=2$
 Distance = $(\ln 2 - \frac{1}{2}) + (2 - \ln 5) - (\frac{1}{2} - \ln 2)$
 $= 1 + 2\ln 2 - \ln 5$ metres
 $\approx 1 - \ln \frac{5}{4}$

b) $P = r + r + r\theta = 20$
 $\theta = \frac{20-2r}{r}$
 $A = \frac{1}{2}r^2\theta = \frac{1}{2}r^2(\frac{20-2r}{r})$
 $= r(10-r)$
 $A = 10r - r^2$

(106)

dr
 max when $\frac{dA}{dr} = 0$ $10 - 2r = 0$
 $r = 5$
 Test: $\frac{d^2A}{dr^2} = -2 < 0$ concave down
 $\theta = \frac{20-10}{5} = 2$
 $\theta = 2$
 $\theta = \frac{2.182}{\pi}$
 $= \frac{360}{\pi}$
 $= 114.59$
 $\approx 115^\circ$ (nearest degree)

Q9. $ax^2 + 5x + a = 0$
 $\Delta = 5^2 - 4(a)(a)$
 (i) $\Delta = 25 - 4a^2$
 (ii) neg. def. $\Delta < 0$ and $a < 0$
 $25 - 4a^2 < 0$
 $4a^2 > 25$
 $a < -\frac{5}{2}$ or $a > \frac{5}{2}$
 but $a < 0$
 $\therefore a < -\frac{5}{2}$

b) $\log_x 4 = \log_3 \sqrt{3}$
 $= \frac{1}{2} \log_3 3$
 $\therefore \log_x 4 = \frac{1}{2}$
 $x^{\frac{1}{2}} = 4$
 $x = 16$

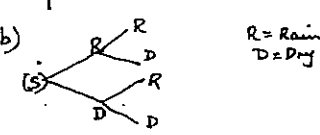
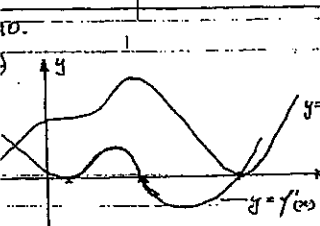
P.T.O.

cont.
 i) $S = Ae^{kt}$
 $5000 = Ae^{0.07 \times 8}$
 $A = 5000$

$\frac{dS}{dt} = kAe^{kt}$
 $0 = k(5000)e^{0.07 \times 8}$
 $k = \frac{350}{5000}$
 $= 0.07$

$S = 5000e^{0.07 \times 8}$
 $= 5000e^{0.56}$
 $= 5362.54$
 ≈ 5363 (to the nearest whole number)
 8753 ants.

i) $\alpha = \sin^2 x$ $1 - \alpha = \cos^2 x$
 $S_{\alpha} = \frac{\alpha}{1-\alpha} = \frac{\sin^2 x}{1-\sin^2 x}$
 $= \frac{\sin^2 x}{\cos^2 x} = \tan^2 x$



b) (ii) $P(\text{at least one in } n \text{ days}) = 1 - (0.85)^n$

(iii) $1 - (0.85)^n \geq \frac{99}{100}$
 $(0.85)^n \leq \frac{1}{100}$
 $n \ln 0.85 \leq \ln \frac{1}{100}$
 $n \geq \frac{\ln 0.01}{\ln 0.85}$
 $\geq 28.336 \dots$
 $\therefore$ least no. of days is 29.

c) $9\% \text{ pa} = 7.5\% \text{ p month}$

ii) let A_n be amount owing at the end of n months.
 $A_1 = P(1.0075) - 600$
 $A_2 = A_1(1.0075) - 600$
 $= P(1.0075)^2 - 600(1.0075) - 600$

$A_n = P(1.0075)^n - 600(1.0075)^{n-1} - 600(1.0075)^{n-2} - \dots - 600(1.0075) - 600$
 $= P(1.0075)^n - 600(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1})$
 $= P(1.0075)^n - 600 \left(\frac{1.0075^n - 1}{1.0075 - 1} \right)$

$A_{120} = P(1.0075)^{120} - 600 \left(\frac{1.0075^{120} - 1}{0.0075} \right)$
 $\frac{1}{2}P = P(1.0075)^{120} - 600 \left(\frac{1.0075^{120} - 1}{0.0075} \right)$
 $P(1.0075^{120} - 5) = 600 \left(\frac{1.0075^{120} - 1}{0.0075} \right)$
 $P = \frac{600(1.0075^{120} - 5)}{(1.0075^{120} - 5)(0.0075)}$
 $= 59,501.45$

(107)

9. cont.

i) $S = Ae^{kt}$

$5000 = Ae^0$

$A = 5000$ ✓

ii) $\frac{dS}{dt} = kAe^{kt}$

$350 = k(5000)e^0$ ✓

$k = \frac{350}{5000}$

$= 0.07$ ✓

i) $S = 5000e^{0.07 \times 8}$

$= 5000e^{0.56}$

$= 5362.54$

$= 5363$ (to the nearest whole no)

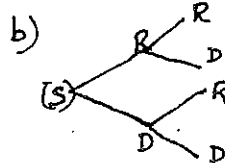
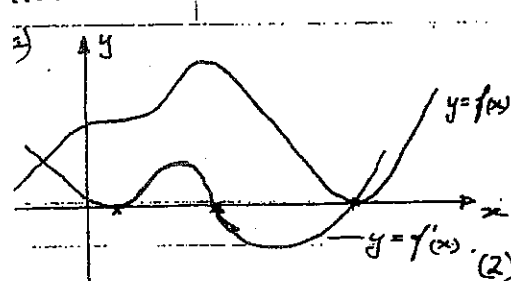
8753 auto.

ii) $a = \sin^2 x$ $r = \sin^2 x$

$S_{\infty} = \frac{a}{1-r} = \frac{\sin^2 x}{1-\sin^2 x}$ ✓

$= \frac{\sin^2 x}{\cos^2 x} = \tan^2 x$ ✓

no.



R = Rain
D = Dry

b) ii) $P(\text{at least one in } n \text{ days}) = 1 - (0.85)^n$

(iii) $1 - (0.85)^n \geq \frac{99}{100}$

$(0.85)^n \leq \frac{1}{100}$

$n \ln 0.85 \leq \ln 0.01$

$n \geq \frac{\ln 0.01}{\ln 0.85}$

$\geq 28.336..$

$\therefore$ least no. of days is 29.

c) $9\% \text{ pa} = 75\% \text{ p month}$

ii) let A_n be amount owing at the end of n months.

$A_1 = P(1.0075) - 600$

$A_2 = A_1(1.0075) - 600$

$= P(1.0075)^2 - 600(1.0075) - 600$

$A_n = P(1.0075)^n - 600(1.0075)^{n-1} - 600(1.0075)^{n-2} \dots$
 $- 600(1.0075) - 600$

$= P(1.0075)^n - 600(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1})$

$= P(1.0075)^n - 600 \left(\frac{1.0075^n - 1}{1.0075 - 1} \right)$

$A_{120} = P(1.0075)^{120} - 600 \left(\frac{1.0075^{120} - 1}{0.0075} \right)$

$\frac{1}{2}P = P(1.0075)^{120} - 600 \left(\frac{1.0075^{120} - 1}{0.0075} \right)$

$P(1.0075^{120} - 0.5) = 600 \left(\frac{1.0075^{120} - 1}{0.0075} \right)$

$P = \frac{600(1.0075^{120} - 1)}{(1.0075^{120} - 0.5)(0.0075)}$

$= 59,501.45$

(107)